Type MS · Draw a bar chart and justify its use

Answer 1 of 8 · Question 5 part b.i · 5 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
5	(a)	• change in base sequence; • causes change in protein synthesis; • causes change in phenotype;		3
	(b) (i)	• axes labelled with units; • suitable scale; • correct plots; (3 x 2) • bars of equal width;		5
	(ii)	• 43 000 × 0.0015; • = 64.5; • 65;		3
	(c)	• extra chromosome at pair 21; • diploid number 47; • female because 2 X chromosomes; • at pair 23;		4

Total : 15 marks

Type MS · Draw a bar chart and justify its use

Answer 1 of 8 · Question 5 part b.i · 5 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a bar chart and justify its use

Answer 2 of 8 · Question 4 part b.i · 3 marks

2020 NOV · Paper 01 (regional R) · find part b.i in this mark scheme extract

4 (a)	• ankles(1) • protein(1)		2
(b) (i)	• axes labels(1) • axes scales(1) • correctly plotted bars(1)	Max 2 marks for line graph	3
(ii)	improved/varied/balanced diet/education (in nutrition) / international aid/less protein in diet in 2010(1)		1
(iii)	• 95 ÷ 1978 (x 100) (1) • 4.8(%)	Full marks for correct final answer	2

Total for Question 4 = 8 marks

Type MS · Draw a bar chart and justify its use

Answer 3 of 8 · Question 3 part a.i · 5 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
3	(a) (i)	• correct labelling of axes (1) • suitable scale (1) • X and Y axis correct way round (1) • plotting (2)	minus 1 for each incorrect plot	5
	(ii)	Any two from • few at lower end/150-151/top end/160 (1) • majority in middle/154-157 (1) • increases to 154-155 then decreases (1)		2
	(b)	Any three from • nutrition (1) • sex (1) • age (1) • genes/genetically based (1)		3

Total 10 marks

Type MS · Draw a bar chart and justify its use

Answer 3 of 8 · Question 3 part a.i · 5 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a bar chart and justify its use

Answer 4 of 8 · Question 4 part a.i · 4 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

4 (a) (i)	• suitable scales; • axes labelled; • correct plots;;	Number of teeth on X axis, number of students on Y axis	4
(ii)	• 13; / 30 • x 100; • 43.3(%);	Allow 43(%)	3
(b)	• wisdom teeth/molars not developed; • jaw too small to accommodate extra teeth;		2
(c)(i)	• sugar coats/left in teeth/forms plaque; • bacteria use sugar (for respiration/energy); • convert sugar to lactic acid; • acid dissolves/erodes/breaks down enamel;	Allow feed on Allow release acid	4
(c)(ii)	any two from • regular brushing of teeth/use fluoride toothpaste/flossing; • visit dentist regularly; • add fluoride to tap water		2 Total 15

Type MS · Draw a bar chart and justify its use

Answer 4 of 8 · Question 4 part a.i · 4 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a bar chart and justify its use

Answer 5 of 8 · Question 5 part b.i · 4 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

5 (a) (i)	A = eardrum; B = cochlea;		2
(ii)	Any two from the following • collects sound waves/vibrations/determines direction of sound; • directs them through the auditory / ear canal; • to the eardrum;		2
(b) (i)	• axes labels; • units; • bars correctly plotted; • scale;		4
(ii)	• (20 – 14) 6; • (6 ÷20) -0.30; • (x100) -30.00%	Allow ecf is minus symbol not given (Max 2 marks)	3
(c) (i)	Any two from the following • gender variation; • some people may have an ear defect;		2
(ii)	EITHER • check results; • with (data) from another group; OR		2

Type MS · Draw a bar chart and justify its use

Answer 5 of 8 · Question 5 part b.i · 4 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

	• repeat the experiment; • calculate mean;		
			Total 15

Question number	Answer	Notes	Marks

Type MS · Draw a bar chart and justify its use

Answer 6 of 8 · Question 2 part a.i · 4 marks

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

2 (a)	(i)	• Y axis label and units; • X axis bars identified; • scale; • plots;	line graph score mpt 1-3	(4)
	(ii)	1;		(1)
(b)		In the following order: • lag; • exponential; • equal to; • death; • less than; • dying;		(6)

Total for Question 2 = 11 Marks

Type MS · Draw a bar chart and justify its use

Answer 7 of 8 · Question 3 part a.i · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
3 (a)	(i)	• time in minutes; • 0-10, 11-20, 21-30, 31-40,41-50,51-60;		2
	(ii)	• 75 + 72 + 68; plus/minus 2 for each one • 215 cm^3;	ECF	2
	(iii)	• rapid reaction/a lot/large amount of gas produced, at start; • rate slows/less gas (for next 10 minutes); • drops a lot/little gas produced (after 40 minutes);		3
	(iv)	any three from • at start plenty of substrate/many E-S complexes formed; • substrate used up/less substrate available; • reaction slows/less gas/product, produced/less enzyme action; • most substrate used at end so little gas produced;		3
	(b)	• less gas produced/reaction lower/less; • because active sites blocked;		2

Total 12

Type MS · Draw a bar chart and justify its use

Answer 7 of 8 · Question 3 part a.i · 2 marks

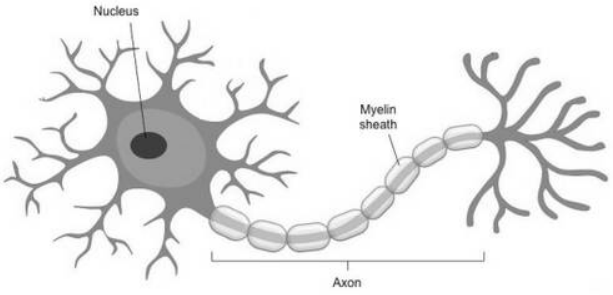
2023 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a bar chart and justify its use

Answer 8 of 8 · Question 3 part b.iii · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

3	(a)	(i)		1 mark for diagram 3 marks for labels	4
		(ii)	• transmits electrical/nerve impulses; • from CNS/brain/spinal cord/relay neurone, to effector organs/named effector organ;		2
	(b)	(i)	Any six from 1 (student X) holds up the ruler between the thumb and forefinger (of other student); 2 then drops/releases the ruler; 3 (other) student catches the ruler; 4 record the reading on the ruler/distance fallen when caught; 5 repeat for <u>this</u> student; 6 using the same distance between thumb and forefinger; 7 find mean/average value of reading on ruler; 8 calculate reaction time (using formula); 9 repeat with <u>other</u> students;		6
		(ii)	the reading on the ruler/ reaction time;		1
		(iii)	independent variable not continuous/is discrete;		1
		(iv)	reaction time = √(0.15 ÷ 5); reaction time = 0.17 (s);	accept 0.173(2)	2
				Total for Question 3 = 16 marks	